

A Morphologically-Aware Transformer Network for High-Fidelity V1-V5 ECG Lead Generation

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Abstract— Cardiovascular disease (CVD) remains the leading cause of death worldwide. 12-lead electrocardiograms (ECGs) are critical for comprehensive CVD diagnosis but require specialized equipment to be recorded. Synthesizing a 12-lead ECG using a subset of leads is highly valuable but is challenging due to the intricacies and variation inside an ECG signal. This paper introduces a multi-stage transformer architecture that utilizes only three leads (Ie. I, III, V6) to synthesize the precordial leads V1 – V5, which are otherwise difficult to record outside clinical environments. The proposed model implements phase-alignment and multi-resolution filtering to fuse segment-specific feature maps to enhance diagnostically critical parts of each of the ECG segments. A mixture-of-experts decoder along with an adaptive multi-component loss function is used to capture and synthesize the subtle morphological attributes of an ECG (ie. P, QRS, T-waves) supporting high-fidelity ECG synthesis. Results show a very high qualitative correlation between synthesized ECG and original ECGs for both normal and abnormal conditions and also achieve a low Average Absolute Error (0.221) and Normalized Root Mean Squared Error (0.052).

Keywords—Neural networks, generative models, multi-scale models, transformers, convolutional networks, electrocardiograms

I. INTRODUCTION

A standard 12-lead ECG consists of six limb leads (I, II, III, aVR, aVL, aVF), which capture electrical activity in the frontal plane, and six precordial leads (V1-V6), which cover the horizontal plane, coming together to form a three-dimensional representation of the electrical activity of the heart [1,2]. Smart wearable devices commonly can record or compute ECGs for the limb leads; however, precordial leads are more clinically demanding to record, allowing various types of CVD to go undetected, leading to long-term cardiac damage [3]. This underscores the necessity for new computational methods to generate the precordial leads. Each ECG heartbeat comprises three main features: the P-wave (atrial depolarization), the QRS complex (ventricular depolarization), and the T-wave (ventricular repolarization). Cardiologists diagnose CVD by analyzing morphological patterns and intricacies of these waves, such as the gentle curve of the P-wave, the sharp peaks of the QRS complex, and the parts in between these segments [4]. The significant variability in ECG recordings and the combination of high- and low-frequency components within ECG features pose a significant challenge for the synthesis of a high-fidelity ECG. This study proposes a transformer-based architecture for ECG

lead synthesis of leads V1-V5 from a limited lead set. Leads I, III, and V6 serve as the input leads from which the precordial leads V1-V5 are generated. The remaining limb leads (II, aVR, aVL, aVF) can be computed directly from the input leads using Einthoven's and Goldberger's equations [5,6,7]. Integration of these three methodologies—Einthoven's formulae, Goldberger's formulae, and the proposed transformer-based network—yields a complete 12-lead ECG from the limited input.

II. METHODS

A. Data Preparation

The complete 12-lead ECGs for model training and evaluation were from the PTB-XL [8] dataset, comprising of more than 21,799 ECGs. While the PTB-XL dataset is composed of a large number of ECGs it is recorded from a German clinical population limiting demographic, ethnic and geographic diversity. This study utilized a subset of 15,190 ECGs, split between five cardiovascular conditions: Normal, Anteroseptal Myocardial Infarction (ASMI), Inferior Myocardial Infarction (IMI), Left Anterior Fascicular Block (LAFB), and Left Ventricular Hypertrophy (LVH). Single beats were extracted from the full ECG recording (about 10 beats) to construct an dataset with an even distribution for each disease type, allocating 4,500 single beat ECG recordings for each type. The final dataset comprised of 22,500. The dataset was partitioned into training and testing sets using a 70%-30% split. Probabilistic data augmentation techniques consisting of amplitude scaling, temporal shifting, and additive noise were used to enable scale-invariant learning, resulting in stable model training. The augmentation probability was set to 0.4.

B. Model Overview

This paper proposes a region-aware transformer network to synthesize leads V1-V5 from the input leads (I, III, V6). The framework specifically captures and reconstructs the key morphological features of an ECG (P, QRS, and T-waves) by integrating regional feature extractors employing custom attention mechanisms to precisely focus on these distinct segments. The proposed architecture comprises five key components:

- (1) Morphological Feature Extractor (MFE) uses multi-resolution convolutions with dual saliency gates to capture global, local and P-wave ECG features

- (2) Multi-scale Feature Module (MSF) aggregates features across fine, medium and coarse temporal scales
- (3) Region-aware Transformer Module (RTM) utilizes multi-resolution convolutional and recurrent layers to extract features of from each ECG region
- (4) Global Statistical Aggregator (GSA) distills the temporal feature maps into concise descriptors for each region
- (5) Morphology Preserving Decoder (MPD) consists of a sub-decoder network to synthesize the distinct segments of the waveform of the target lead

Each lead is synthesized using a separate model trained for that specific lead, using an identical architectural framework with individually tuned hyperparameters. Additionally, a custom adaptive loss function was developed to provide comprehensive feedback during training. In the subsequent sections, let $X \in \mathbb{R}^{B \times T}$ where (B) denotes batch, (T) represents time samples, and (L) indicate the input leads. The target lead is denoted by $y \in \mathbb{R}^{B \times T}$.

C. Morphological Feature Extractor (MFE)

The MFE is designed to extract features from the ECG signals at multiple temporal resolutions using simultaneous convolutional streams, each utilizing distinct kernel widths. To allow the network to model global and local structures in the ECG (atrial and ventricular complexes), these streams are then adaptively fused and gated together. This approach generates a comprehensive set of features that represent all parts of the ECG. Additionally, the MFE consists of two independent saliency gates: one gate for global features and another to explicitly target the subtle characteristics of the P-wave. These gates afford the model more fine grained control over which output channels enter the attention mechanism—a feature absent from standard convolutional layers.

The input data is first processed using depth-wise separable convolutional layers with three kernel sizes of $k \in \{3, 5, 7, 9, 11, 15, 31\}$ which each capturing short-, mid- and long-span dependencies with $Z^{(s)} = \text{ReLU}(\text{DWConv}_{k_s}(X) + b_s)$. A bi-directional LSTM further encodes the temporal information from the fine stream. The channels are concatenated into H_t . The local saliency is then encoded through a gated mask of $g_t = \sigma(W_t H_t + b_g)$, $\tilde{H}_t = g_t \odot H_t$ which spots locally important channels and “brightens” them while “dimming” unimportant ones, helping recognize and enhance the subtle morphological features of the ECG waves, especially the P-wave. Ventricular and baseline filters are appended yielding M_t . A second

attention gate of $\sigma(W_m M_t + b_m) \odot M_t$ reweights M_t into Z_t . This couples the raw lead data along with the emphasized morphology features, supplying a rich feature set for the subsequent stages.

D. Multi-scale Feature Module (MSF)

Using feature maps outputted by the MFE, the Multi-scale Feature Module (MSF) transforms them into fine, medium, and coarse grained streams, each processed using with different strides and kernel sizes. Subsequently, the MSF adaptively learns to reassemble these distinct channels into a single tensor, enriching the data representation with both local fluctuations and low-frequency features. After this filtering step, shorter sequences are interpolated back to their original sample length. This system utilizes a “trim-and-repeat” alignment, which ensures coherence and minimizes aliasing artifacts that are common in other up-sampling techniques.

The MSF distributes Z across three different stride lengths of $r \in \{1, 2, 3\}$ where r is the original rate. Depth wise convolutions are applied to each channel of Z with each channel using different strides which are then mapped into a new feature space producing $\tilde{C}^{(r)} = \text{Repeat}_r(C^{(r)})_{:T}$. The repetition produces phase-aligned tensors with the fine grid by cropping after repetition. Finally, $V_t = \text{BN}(\phi(W_f[C_t^{(1)}; \tilde{C}_t^{(2)}; \tilde{C}_t^{(3)}; U_t]))$ where U_t is a 1×1 temporal projection. V_t represents fused, coarse- and fine-grained features without interpolation artifacts.

E. Region Aware Transformer Module (RTM)

The RTM processes the multi-scale tensor of V_t with a customized self-attention mechanism that uses explicit biases toward specific ECG phases. This customization allows the network to effectively capture early-sample features, such as the P-wave, while simultaneously maintaining long-range dependencies, yielding a more informative global representation than when compared to uniform attention mechanisms.

The global rhythm is modeled by six identical transformer layers. Prior to attention a sinusoidal term, $II_{t,d} = \sin(t/10^{4d/192})$, is injected with a amplitude of 0.1. For each head of h , $\alpha_{t,s}^{(h)} = \frac{Q_t^{(h)} K_s^{(h)T}}{\sqrt{d_h}} + \log(1.5) \mathbf{1}_{s \leq 15}$, where the additive bias is applied selectively to P-wave positions. After *softmax* normalization, the gated residual pair $\hat{V}_t = \text{LN}(V_t + g_t^{(1)} \odot \text{MHA}(V_t))$, $\bar{V}_t = \text{LN}(\hat{V}_t + g_t^{(2)} \odot \text{FFN}(\hat{V}_t))$ assures that only diagnostically relevant features are pushed forward,

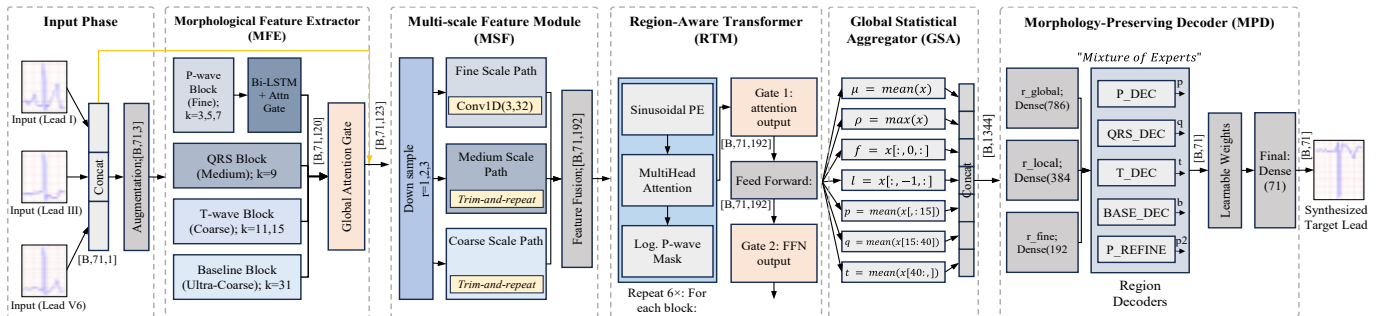


Figure 1. Proposed transformer model diagram illustrating region-aware methodology used to synthesize an ECG from a three-lead input.

concentrating computation on the areas that cardiologists naturally focus.

F. Global Statistical Aggregator (GSA)

The GSA integrates temporal sequences into a holistic feature vector by stacking seven complementary statistical descriptors. This strategy retains extrema and region specific averages that would traditionally dissipate under single-pool methods. This richer summary ensures low-energy segments, such as the ST segments, are decoded accurately.

The GSA concatenates seven distinct descriptors: $g = [\mu; \gamma; \delta_0; \delta_{70}; \alpha; \nu; \tau]$, where μ and γ are the temporal mean and max of $\tilde{V}$, δ_0 and δ_{70} are the first and last hidden states, and α, ν, τ are the interval means over atrial and ventricular and de-/repolarization segments. This unifies global shapes and region-specific cues where g is a highly informative image of the entire ECG signal.

G. Morphology-Preserving Decoder (MPD)

The MPD synthesizes the target lead using a mixture-of-experts decoder with five parallel linear experts, each specializing in different waveform families, which are combined softly via a learned gating matrix. This circumvents the limitations of simple decoders and preserves the delicate slope of a P-wave while simultaneously reproducing high-amplitude QRS spikes. Each of these experts are implicitly supervised via the global loss, achieving region specific accuracy without the need for manual masks.

A three-level projection refined g to $h^{(3)}$ where the five linear “experts” each specialize in distinct waveform families: $\hat{y}^{(j)} = W^{(j)}h^{(3)} + b^{(j)}, j = 1 \dots 5$. The outputs are then softly ensembled via $\omega = \text{softmax}(W_\omega h^{(3)} + b_\omega), \tilde{y} = \sum_{j=1}^5 \omega_j \hat{y}^{(j)}$, finally, producing $\hat{y} = W_f \tilde{y} + b_f$. These formulae permit the decoder to adaptively privilege and weight the expert whose morphology best represents the current segment while still maintaining a global linear representation. This yields crisp P-waves and sharp QRS complexes.

H. Loss Function & Model Training

The loss function for the transformer model is designed to reward both global and region-specific waveform fidelity, overcoming the limitations of standard uniform loss functions that often fail to capture the morphological features of ECG signals. Instead, the custom loss function blends various types penalties to accurately represent the ECG signals. Where $\hat{y}$ denotes the predicted and reference beats, the total cost is $\text{Loss} = \beta(0.20\text{MSE} + 0.15\text{Grad}_1 + 0.10\text{Grad}_2 + 0.25\text{Seg}_{\text{ATR}} + 0.15\text{SegGrad}_{\text{ATR}} + 0.05\text{Peak} + 0.03\text{Corr} + 0.02\text{FFT})$ combines eight orthogonal terms: point-wise voltage error (MSE); first- and second-order derivative errors ($\text{Grad}_1, \text{Grad}_2$); interval specific penalties on the atrial segment ($\text{Seg}_{\text{ATR}}, \text{SegGrad}_{\text{ATR}}$); an extremum-sensitive peak term; a beat normalized correlation term; and a spectral magnitude error

(FFT). The additive factor $\beta = \text{clip}(\frac{\|y\|_2^2}{BT}, 0.1, 2.0)$ scales the loss to fit the signal energy to prevent over and under penalization on low and high amplitude beats respectively. All

models for the different leads were trained using the Adam optimizer with global gradient norm clipping set to 1.0. A ReduceLROnPlateau scheduler was employed to adaptively adjust the learning rate based on validation loss.

III. RESULTS

The proposed transformer network is evaluated using both quantitative and qualitative methods. Quantitatively, performance is assessed with Average Absolute Error (AAE; Table 1), which measures the mean deviation in millivolts, and Normalized Root Mean Squared Error (NRMSE; Table 2), a scaled metric enabling comparisons across leads and conditions. Qualitatively, as illustrated in Fig. 2, overlays of synthesized and real ECGs show accurate reproduction of key morphological features—including the P-wave, QRS complex, and T-wave—across cardiovascular conditions.

IV. CONCLUSION

The proposed model was designed to focus on the clinically significant features of an ECG with the goal of making the synthesized ECG leads of diagnostic quality. The results show that this approach is able to synthesize ECGs that are morphologically similar, with low errors when compared to the real ECG, indicating that the model is able to capture the subtle variations that are clinically significant across all the subsegments of the ECG (ie. P, QRS, T). The use of a multi-stage architecture and a learned multi-component loss function was shown to be an effective method to capture heterogenous attributes associated with both morphology and timing, making it suitable for both automated diagnosis and ECG synthesis.

Table 1. Average Absolute Error for Leads V1-V5

Leads	AAE of Synthesized Leads V1-V5					
	Normal	ASMI	LAFB	LVH	IMI	Average
Lead V1	0.2387	0.2243	0.2879	0.1851	0.2462	0.2365
Lead V2	0.2346	0.2738	0.2703	0.2448	0.2639	0.2575
Lead V3	0.2334	0.3072	0.2238	0.2516	0.2556	0.2543
Lead V4	0.1623	0.2961	0.1880	0.1862	0.2113	0.2088
Lead V5	0.1170	0.2092	0.1593	0.1102	0.1401	0.1472

Table 2. Normalized Root Mean Squared Error for Leads V1-V5

Leads	NRMSE of Synthesized Leads V1-V5					
	Normal	ASMI	LAFB	LVH	IMI	Average
Lead V1	0.0535	0.0548	0.0684	0.0427	0.0592	0.0557
Lead V2	0.0600	0.0749	0.0676	0.0619	0.0648	0.0659
Lead V3	0.0562	0.0862	0.0547	0.0604	0.0595	0.0634
Lead V4	0.0320	0.0733	0.0403	0.0399	0.0435	0.0458
Lead V5	0.0216	0.0455	0.0304	0.0220	0.0265	0.0292

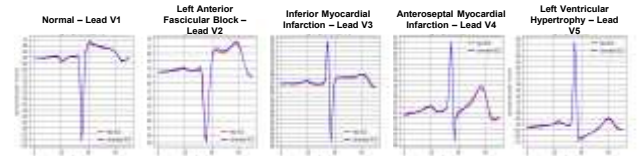


Figure 2. Overlaid synthesized (blue) and actual (red) ECGs across five conditions.

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